

Urological Oncology Mutations

Relevance to the SOLANGE Quantum Simulation Platform

1. Background -- The Non-Druggable Problem in Urology

Targeted oncology therapy depends on the existence of a druggable binding pocket -- a defined three-dimensional site on the mutant protein to which a small molecule can bind with sufficient affinity to alter protein function. For a significant subset of urological cancer mutations, no such pocket exists. These mutations cause loss-of-function in tumour suppressor proteins whose conformational consequences cannot be addressed by classical docking or small-molecule design. The patient carries a Tier I pathogenic variant for which current oncology has no pharmacological answer.

The SOLANGE platform was designed specifically for this class of mutation. It applies quantum simulation -- the only method capable of characterising electronic structure at chemical accuracy for active spaces beyond the classical chemistry ceiling (~18 active electrons under CCSD(T)) -- to generate ground-state energetics that form the scientific basis for future small-molecule design.

2. Renal Cell Carcinoma (RCC) -- Mutation Landscape

Clear cell RCC is the most common renal malignancy (~75% of cases). Its mutational landscape divides sharply into oncogenic targets amenable to approved therapies, and tumour suppressor loss-of-function mutations for which no targeted therapy exists.

Gene	Role	Mutation type	Approved therapy	SOLANGE-relevant
FGFR3	Oncogene	Activating	Erdaftinib (FDA 2019)	NO
MTOR	Oncogene	Activating	Everolimus / Temsirolimus	NO
VHL	Tumour suppressor	Loss of function	Belzutifan (targets HIF-2a, not VHL itself)	INDIRECT
PBRM1	Chromatin remod.	Loss of function	None	YES
BAP1	Deubiquitinase	Loss of function	None	YES
SETD2	Histone MeTase	Loss of function	None	YES
KDM5C	Histone DeMase	Loss of function	None	YES

- > PBRM1, BAP1, SETD2 and KDM5C together account for ~40-60% of clear cell RCC. All are loss-of-function tumour suppressors with no druggable pocket -- the primary target class for SOLANGE in renal oncology.

3. Bladder Cancer -- Mutation Landscape

Urothelial carcinoma presents a similarly bifurcated landscape. FGFR3 alterations have been successfully targeted; a parallel class of tumour suppressor mutations remains pharmacologically inaccessible.

Gene	Role	Mutation type	Approved therapy	SOLANGE-relevant
FGFR3	Oncogene	Activating	Erdaftinib (FDA 2019)	NO

PIK3CA	Oncogene	Activating	Investigational (PI3Ki)	NO
ERCC2	DNA repair	Loss of function	Biomarker only (cisplatin)	NO
TP53	Tumour suppressor	Loss of function	None	YES
RB1	Tumour suppressor	Loss of function	None	YES
ARID1A	Chromatin remod.	Loss of function	None	YES
CDKN2A	Tumour suppressor	Loss of function	None (CDK4/6i req. intact RB1)	YES

- > TP53 mutations in bladder cancer are mechanistically identical to TP53 C275F in NSCLC -- the anchor case for which SOLANGE was originally designed. The platform is immediately applicable without architectural modification.

4. Computational Scale -- Why Classical Chemistry Cannot Solve This

Classical quantum chemistry (CCSD(T)) is intractable beyond ~18 active electrons. Every non-druggable mutation listed above involves a binding site with an active space well beyond this ceiling. The table below maps each target to its estimated qubit requirement under Jordan-Wigner encoding (2 qubits per electron).

Gene / Mutation	Active-site model	Est. active e-	Est. qubits	Platform phase
TP53 C275F	L3 loop + Zn coordination residues	~44e	~88q	3B -- QPU
BAP1 C91S	UCH catalytic triad (3 residues)	~18-30e	~36-60q	3A / 3B
PBRM1 LOF	BAH domain / bromodomain	~24-40e	~48-80q	3A / 3B
SETD2 LOF	SET catalytic domain	~20-36e	~40-72q	3A / 3B
RB1 LOF	A/B pocket domain	~30-50e	~60-100q	3B -- QPU
ARID1A LOF	DNA-binding domain	~28-44e	~56-88q	3B -- QPU

- > Electron counts are structural estimates. BAP1 C91S sidechain model computation completed (CASSCF/STO-3G, June 2026). Full catalytic-triad calculation pending PDB structure access.

5. What SOLANGE Does and Does Not Do

- DOES:** Characterises the electronic ground-state of a non-druggable mutant binding site at quantum-chemical accuracy, under FDA 21 CFR Part 11 audit trail (P1-P9).
- DOES:** Generates binding-energy rankings for candidate molecules against a specific mutant conformation -- precision medicine at the individual-patient mutation level.
- DOES:** Produces reproducible, cryptographically sealed simulation records suitable for regulatory submission and research publication.
- DOES NOT:** Replace clinical judgment or oncology treatment protocols.
- DOES NOT:** Create approved drug candidates -- it generates computational evidence to guide drug discovery programmes.
- DOES NOT:** Operate on mutations for which approved targeted therapy already exists (FGFR3, MTOR, VEGFR-pathway targets are outside scope).

6. A Message to Urological Oncologists

You already know the hardest cases.

A patient arrives with a genomic report. The variant is Tier I -- pathogenic, confirmed. But the affected protein is a tumour suppressor with no known binding pocket, no approved targeted therapy, and no clinical trial addressing this mutation. The report is genomically actionable and therapeutically inaccessible. You have seen this before.

SOLANGE was built for this exact gap.

It does not promise a treatment today. What it does is characterise -- at quantum-mechanical precision, with a complete regulatory audit trail -- the electronic structure of that specific mutant site. That characterisation is the prerequisite for any future small-molecule design targeting this protein. Without it, drug discovery cannot begin.

Every non-druggable mutation you document is a case that matters. The platform is ready.